

ECE NUR BAYRAKTAR

COMPUTER ENGINEERING STUDENT

CONTACT

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EDUCATION

Dokuz Eylul University

B.Sc. Computer Engineering, 3rd Year

Hasan Turek Anatolian High School

Graduated 2nd in class, 2022

TECHNICAL SKILLS

Languages: Python, Java, C, C#, SQL, JavaScript, TypeScript, HTML, CSS

AI & Retrieval: RAG, local LLMs, Phi-3.5 Mini, Foundry Local, prompt engineering, embeddings, BM25

ML & Data: TensorFlow, Scikit-learn, Sentence Transformers, NumPy, preprocessing, evaluation

Embedded AI: STM32, STM32CubeMX, HAL, UART, ADC, X-CUBE-AI

Applications: PySide6, React Native, Expo, Spring Boot, .NET, REST APIs

Data & Tools: PostgreSQL, MySQL, MSSQL, Docker, Git, regression testing

LANGUAGES

English - Advanced
German - A2-B1

INTERNATIONAL EXPERIENCE

Erasmus+ Exchange Program, Poland

Participated in a high-school exchange program, gaining early international and cross-cultural experience.

PROFESSIONAL SUMMARY

Curious, disciplined, and growth-oriented Computer Engineering student who enjoys understanding how complex systems work and turning ideas into reliable, user-focused solutions. I value thoughtful problem solving, clear communication, and continuous improvement, and I am eager to contribute within collaborative engineering teams while deepening my expertise in AI and software development. Selected work and technical case studies are available in my [portfolio](#).

WORK EXPERIENCE

AI Intern

Microsoft Summer Internship Program | Summer 2026

- Developed a retrieval-augmented generation (RAG) application powered by a locally hosted large language model.
- Built a local RAG pipeline involving document chunking, embedding generation, semantic retrieval, prompt construction, and response generation.

AI & Software Development Intern

Niso Software Technologies Limited Company | 2026

- Contributed to AI-focused development while applying machine learning and software engineering skills in a professional environment.
- Applied data processing, model development, and software engineering practices to production-oriented AI work.

Software Development Intern

Niso Software Technologies Limited Company | 2025 | 25-day internship

- Developed a machine learning-based command recognition application designed for deployment on an STM32 microcontroller.
- Combined data processing and model development with embedded deployment constraints.

PROJECTS

Local RAG Assistant - Private Document AI

Python, PySide6, Foundry Local, Phi-3.5 Mini, Sentence Transformers, BM25, PyMuPDF

- Built a privacy-first Windows desktop application and product website for English/Turkish, multi-document question answering with page-level citations and fully local processing after model setup.
- Combined semantic embeddings and BM25 retrieval; added prompt grounding, source inspection, evaluation scripts, and regression tests for measurable answer quality.
- Initially targets banking policies, procedures, regulations, and internal knowledge, with an architecture designed to expand into role-specific assistants across other document-heavy domains.

BloomBook - Book Recommendation Platform

Spring Boot, PostgreSQL, Python, Docker, REST API

- Built and containerized a full-stack recommendation platform with PostgreSQL persistence and a separate Python recommendation service.

STM32-Based Command Recognition Application

STM32, STM32CubeMX, C, HAL, UART, ADC, ML, X-CUBE-AI

- Configured GPIO, UART, and ADC peripherals with STM32CubeMX and HAL, using serial output and LED feedback to validate the embedded data pipeline.
- Prepared command/speech samples, explored frequency-domain feature extraction and compared lightweight CNN and Random Forest approaches under limited-memory constraints.
- Converted the trained model with X-CUBE-AI and integrated generated C code into STM32 firmware for on-device inference experiments.

Dort Bagla Mobile - Realtime Connect Four

React Native, Expo, TypeScript, Firebase Realtime Database

- Implemented local, online-room, messaging, and offline AI modes with synchronized gameplay and Android builds.

Additional Software Projects

Spring Boot, .NET, Java, C, XML, XSD, Databases

- Created layered web applications, CRUD systems, console games, data structures, and schema-validated XML workflows.

CERTIFICATIONS

AI Innovators Internship Program

Microsoft | Aug 2026

Machine Learning Explainability

Kaggle | Aug 2026

Generative AI for Software Development

DeepLearning.AI | Aug 2026

Deep Learning Specialization

DeepLearning.AI | Aug 2026

Automotive Camp

OSD | Jul 2026

Migros Good Future Youth Meeting

Migros | Jun 2026

Occupational Health and Safety

YTNK TV | Feb 2026

C# OOP Fundamentals: Layered Architecture

Udemy | Aug 2025

Job Simulation Certificate

Ludi | Jul 2025

AI Applications: Machine Learning to Deep

Learning

BTK Akademi